

An Explainable Single-Step Neural PPO Framework for AI-Native 6G RAN Scheduling

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Abstract—AI-native radio access network (RAN) operation couples scheduling and radio-resource-management (RRM) decisions across carrier aggregation, spectrum, MIMO, load, mobility, fairness, and cell-edge performance, so an improvement in one key performance indicator (KPI) may require accepting a bounded reduction in another. This paper describes RANOpt AI Beta v17.8, a single-step Proximal Policy Optimization (PPO) implementation for AI-assisted RAN scheduling and RRM studies. The update replaces the earlier bounded action-weight continuation rule with trainable policy and value networks, each using two 64-node hidden layers, so that policy and value gradients propagate through both hidden layers during training. Each training sample is treated as a one-step contextual decision rather than a sequential Markov decision process. A multi-objective reward combines normalised throughput, Jain fairness, cell-edge rate, and a dimensionless resource and energy-efficiency index. For a representative Urban Macro FDD-plus-TDD carrier-aggregation point, the retained deterministic reference study shows the Balanced policy raising modelled cell-edge downlink rate by 16% and aggregate and fairness KPIs by about 8% relative to a Proportional Fair baseline, with the energy-efficiency index improving by 4%. These figures are simulator reference outputs, not measurements or a fresh multi-seed campaign. The contribution is an auditable single-step neural PPO mechanism with an explicitly bounded scope.

Index Terms—6G, RAN, reinforcement learning, PPO, single-step RL, neural network, backpropagation, clipped surrogate objective, MAC scheduling, RRM, energy efficiency

I. INTRODUCTION

AI-native RAN operation introduces a coupled decision space across carrier aggregation, FDD and TDD spectrum, MIMO, traffic load, mobility, fairness, cell-edge performance, resource utilisation, and normalised resource and energy efficiency [1]. The medium access control (MAC) scheduler is a useful controlled research target because improving one KPI may require accepting a smaller improvement, or a bounded degradation, in another. A learning agent for this setting must therefore express KPI trade-offs explicitly rather than optimise a single quantity in isolation.

RANOpt AI Beta v17.8 changes the learning path relative to the audited v17.7 platform [2]. In v17.7 the browser-local continuation rule adjusted a small action-weight vector with a scalar baseline and did not retrain the declared hidden layers. In v17.8 the two 64-node hidden layers of the policy and value networks are directly trainable by PPO, so the reward-driven error reshapes the hidden representation rather than only a set

of output preferences. The learning problem remains a one-step contextual decision: each sample supplies a context, the policy selects one RRM action, the simulator returns one reward, and the episode terminates.

This paper makes three contributions. First, it specifies the single-step neural PPO mechanism, including forward propagation, the clipped surrogate objective, and backpropagation through both hidden layers (Sections II–IV). Second, it defines a dimensionless resource and energy-efficiency index as one component of a multi-objective reward and states precisely what that index does and does not measure (Section III). Third, it retains the deterministic reference KPI study for continuity and reports it with an explicit provenance caveat and a validation plan rather than as a fresh implementation result (Sections VI and VII). The morphology-aware synthetic-data generator, checkpoint persistence, carrier data continuation path, deterministic radio comparison, and engineering traceability of the earlier platform are retained.

II. NEURAL POLICY AND VALUE NETWORK WITH BACKPROPAGATION

The v17.8 model receives an 11-feature normalised context vector $\mathbf{s}$ derived from the selected radio scenario. Forward propagation passes $\mathbf{s}$ through two 64-node hidden layers to a policy head and a value head. The policy head produces five logits, and a softmax maps them to probabilities over five bounded RRM and scheduler actions; the value head produces the scalar baseline $V(\mathbf{s})$:

$$\mathbf{h}_1 = f(\mathbf{W}_1 \mathbf{s} + \mathbf{b}_1), \quad \mathbf{h}_2 = f(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2) \quad (1)$$

$$\pi(a | \mathbf{s}) = \text{softmax}(\mathbf{W}_p \mathbf{h}_2 + \mathbf{b}_p), \quad V(\mathbf{s}) = \mathbf{W}_v \mathbf{h}_2 + b_v \quad (2)$$

where f is the hidden activation, $\mathbf{W}_1, \mathbf{W}_2$ and $\mathbf{b}_1, \mathbf{b}_2$ are the hidden weights and biases, and $\mathbf{W}_p, \mathbf{W}_v$ with $\mathbf{b}_p, b_v$ are the policy and value head parameters.

In v17.8 both hidden layers participate directly in PPO learning. After 1 and 2 produce the action probabilities and the value estimate, the selected action is evaluated and the policy and value losses are formed. The loss gradients propagate backward from the two heads through hidden layer 2 and then hidden layer 1, so $\mathbf{W}_2, \mathbf{W}_1$, their biases, and the head weights

are all modified by the optimiser. Unlike the v17.7 continuation mechanism, the hidden layers are not fixed feature-processing layers; their weights learn from the reward-driven PPO error.

Backpropagation does not itself decide whether an RRM action is good. The reward and advantage provide the learning signal, PPO defines a constrained policy objective, differentiation determines how each trainable parameter contributed to that objective, backpropagation applies the chain rule through the network graph, and the optimiser updates the numerical weights and biases according to

$$\mathbf{W}_{\text{new}} = \mathbf{W}_{\text{old}} - \alpha \frac{\partial L}{\partial \mathbf{W}} \quad (3)$$

where α is the learning rate and L is the combined loss.

III. SINGLE-STEP EXPERIENCE, REWARD, AND ENERGY INDEX

For each training sample the environment supplies context $\mathbf{s}$, the policy selects one RRM action a , the simulator evaluates that action, and a scalar reward r is formed. Because the episode terminates after this action, the one-step advantage is

$$A = r - V(\mathbf{s}). \quad (4)$$

A positive A indicates that the selected action performed better than the value baseline predicted, and a negative A indicates worse-than-expected performance.

The multi-objective engineering reward is

$$R = w_T T + w_F F + w_{\text{Edge}} E_{\text{edge}} + w_\eta \eta - P \quad (5)$$

where T , F , E_{edge} , and η denote normalised throughput, Jain fairness, cell-edge performance, and the normalised resource and energy-efficiency index, and P is a penalty reserved for QoS or physical constraint violations. For the Balanced objective the representative weights are $w_T = 0.40$, $w_F = 0.30$, $w_{\text{Edge}} = 0.20$, and $w_\eta = 0.10$, and $P = 0$ in the reported configuration. This construction permits realistic KPI trade-offs rather than requiring every KPI to improve simultaneously; the constraint term is defined but inactive in the reported run.

The resource and energy-efficiency index η is a dimensionless simulator quantity normalised to the AI-OFF Proportional Fair (PF) reference value of 1.000. It represents relative resource and energy efficiency within the modelled RAN operating point and is used as one component of 5. A PPO value of 1.040 therefore denotes a 4% improvement in this normalised index relative to the reference; it is not a direct physical measurement in bits per joule. The current simulator does not include a gNB and UE power-consumption model detailed enough to claim physical energy efficiency in bits per joule. Such a claim would require explicit modelling of transmit, RF, baseband, and fixed operating power and is left as future work.

IV. PPO PROBABILITY RATIO AND SURROGATE OBJECTIVE

PPO retains the probability assigned to the sampled action by the old policy and compares it with the probability assigned by the current trainable policy. The probability ratio is

$$\rho(\theta) = \frac{\pi_\theta(a | \mathbf{s})}{\pi_{\theta_{\text{old}}}(a | \mathbf{s})}. \quad (6)$$

A ratio of one means no change in probability, above one means the action became more probable, and below one means it became less probable. The objective is a surrogate because the true expected policy return is not differentiated directly; instead PPO optimises a tractable local proxy built from sampled advantages and the old-to-new probability ratio [3]. This remains a surrogate policy objective even though v17.8 uses a one-action episode:

$$L_{\text{clip}}(\theta) = \mathbb{E}[\min(\rho(\theta)A, \text{clip}(\rho(\theta), 1 - \epsilon, 1 + \epsilon)A)]. \quad (7)$$

Without a constraint, one favourable sample could produce an excessively large policy update. Clipping restricts the surrogate benefit of moving the ratio outside $[1 - \epsilon, 1 + \epsilon]$. For example, with $\epsilon = 0.20$, if an action probability changes from 0.25 under the old policy to 0.35 under the new policy then $\rho = 1.40$, but the clipped positive-advantage surrogate uses at most 1.20. Clipping therefore limits the incentive for a large policy jump; it does not clip the KPI or the reward. The value head is trained toward the observed reward with a value-error term $L_v = (r - V(\mathbf{s}))^2$. The policy term changes action preference and the value term improves the baseline used to estimate advantages; both gradients enter the shared hidden representation.

Through the softmax and policy-head matrix, the loss derivatives with respect to the action logits create an error signal at hidden layer 2. Multiplication by the hidden-activation derivative yields the gradients for $\mathbf{W}_2$ and $\mathbf{b}_2$, and the chain rule then propagates the signal into hidden layer 1 to produce the gradients for $\mathbf{W}_1$ and $\mathbf{b}_1$. The value-loss derivative follows an analogous path from the value head. For a scalar weight w_{ij} the optimiser step is $w_{ij} \leftarrow w_{ij} - \alpha \partial L / \partial w_{ij}$, so with $w_{ij} = 0.120$, $\partial L / \partial w_{ij} = +0.030$, and $\alpha = 0.001$ the plain gradient-descent contribution moves the weight to about 0.11997 before optimiser-specific adaptive scaling. Across many independent one-step samples, these coordinated changes reshape the hidden representation and future action probabilities. This connection of the loss to the full computation graph is the principal distinction from v17.7, whose continuation path could modify five action preferences and a scalar baseline but did not retrain the 64-by-64 hidden matrices. Table I summarises the end-to-end learning path.

V. TRAINING DATA, RADIO CONFIGURATION, AND PERSISTENCE

The workspace retains the morphology-aware synthetic generator. Ten sites with three sectors each produce 30 cells; each cell has 96 records at 15-minute intervals, giving 2,880 base cell-time records over a synthetic 24-hour period.

Table I: End-to-end single-step neural PPO learning path in v17.8.

Stage	v17.8 operation
Input	11 normalised context and state features
Hidden 1	64 trainable nodes; $\mathbf{W}_1, \mathbf{b}_1$
Hidden 2	64 trainable nodes; $\mathbf{W}_2, \mathbf{b}_2$
Policy head	5 logits to softmax action probabilities
Value head	scalar $V(s)$
Learning signal	advantage $A = r - V(s)$
PPO constraint	old-to-new ratio with clipped surrogate
Gradient	backpropagation through heads and both hidden layers
Update	optimiser modifies weights and biases

Expansion into Suburban, Urban, and Dense-Urban variants produces 8,640 morphology-aware rows. The generator seed is 1701 and local training uses seed 12345. Generated columns are correlated through engineering relationships among active UE count, PRB utilisation, SINR, CQI, MCS, throughput, mobility, and buffer or load rather than being independently randomised, with the CQI mapping quantised to 3GPP tables [4]. Explicit PPO training updates the neural checkpoint; normal simulator runs remain inference-only unless training is invoked, and validated carrier CSV records may later be used for continued training.

The controlled study retains a single-cell Urban Macro FDD-plus-TDD carrier-aggregation point, summarised in Table II: NR band n3 FDD at 1800 MHz with 50 MHz bandwidth and NR band n78 TDD at 3.5 GHz with 100 MHz bandwidth, for 150 MHz total; inter-site distance 500 m; 20 active UEs with Medium eMBB traffic; a PF AI-OFF baseline; a 32T32R gNB array; UE maximum downlink and uplink layers of 4 and 2; UE and gNB noise figures of 7 dB and 2 dB; mobility 30 km/h; and the Balanced PPO objective.

Table II: Representative single-cell FDD-plus-TDD carrier-aggregation configuration.

Configuration parameter	Representative value
Morphology and scope	Urban Macro, single cell
Duplexing	FDD + TDD carrier aggregation
Anchor carrier (FDD)	NR n3 (1800 MHz), 50 MHz
Booster carrier (TDD)	NR n78 (3.5 GHz), 100 MHz
Total bandwidth	150 MHz
Inter-site distance	500 m
Active UEs and traffic	20, Medium eMBB
Baseline scheduler	Proportional Fair
gNB antenna	32T32R MIMO
UE max DL/UL layers	4 / 2
Noise figure (UE/gNB)	7 dB / 2 dB
UE mobility	30 km/h
PPO objective and weights	Balanced, 0.40/0.30/0.20/0.10
Generator/training seed	1701 / 12345

VI. REPRESENTATIVE RESULTS AND INTERPRETATION

Table III reports the retained deterministic model-study KPI point for the configuration of Table II. Relative changes follow $100 \times (KPI_{AI} - KPI_{base})/KPI_{base}$ under identical simulated radio conditions. These values are reference outputs carried over for continuity; they are not a fresh post-v17.8 multi-seed measurement. Their purpose here is to show the operating point and the type of multi-objective response being optimised.

The normalised comparison is shown in Fig. 1 and the relative changes in Fig. 2. In this reference point the cell-edge

Table III: Retained deterministic model-study KPI point.

KPI	AI-OFF / PF	PPO Balanced	Change
Cell DL capacity	1.42 Gb/s	1.53 Gb/s	+8.0%
Median UE DL	68.0 Mb/s	73.4 Mb/s	+8.0%
5th-pct UE DL	22.0 Mb/s	25.5 Mb/s	+16.0%
Jain fairness	0.902	0.974	+8.0%
Energy-efficiency index	1.000	1.040	+4.0%

Note: Values are retained reference outputs, not a fresh post-v17.8 multi-seed measurement.

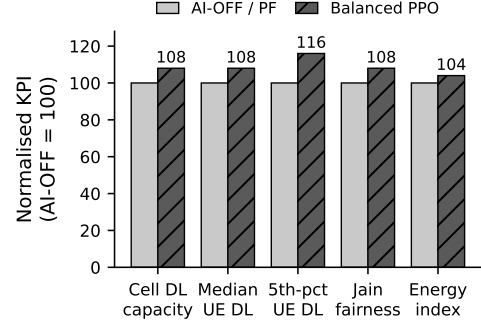


Fig. 1: Normalised AI-OFF/PF versus Balanced PPO KPIs for the FDD-plus-TDD carrier-aggregation point (AI-OFF = 100).

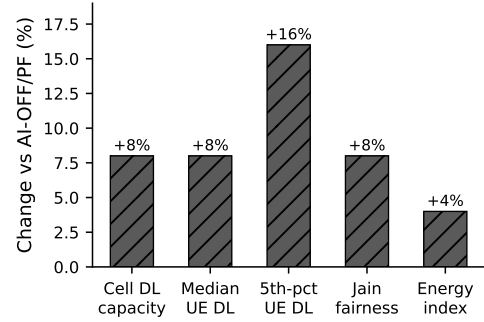


Fig. 2: Relative KPI changes at the retained representative operating point.

(5th-percentile) rate improves more (+16.0%) than aggregate capacity (+8.0%), while fairness rises from 0.902 to 0.974 and the normalised energy-efficiency index improves by 4.0%. This pattern is consistent with the multi-objective reward weighting, which places explicit weight on cell-edge and fairness terms alongside throughput. Because the policy is evaluated in the same simulator that defines its reward, the comparison is in-sample and cannot on its own establish that the effect generalises across loads, seeds, or sites; a fresh paired multi-seed rerun is required before attributing numerical gains to the end-to-end v17.8 neural update.

VII. EXPLAINABILITY, LIMITATIONS, AND FUTURE VALIDATION

Single-step does not mean that only one neural update occurs during training. Many independent samples can be processed and each can update the network; within one episode the sequence is context, one RRM action, one reward, one PPO update, then termination. A future multi-step MDP would carry the post-action state into the next scheduling interval and accumulate rewards across a trajectory, enabling temporal

credit assignment and returns accumulated over the trajectory rather than a single immediate reward [5].

Engineering explainability remains outcome-based rather than assigning physical meaning to individual hidden neurons. A decision trace can expose the configuration identifier, model and checkpoint version, selected action, old and new action probabilities, probability ratio, advantage, clipping status, reward components, and the resulting KPI vector, so that a recommendation can be reproduced and audited under identical deterministic channel parameters.

The main limitations bound the current claims. The built-in data are synthetic engineering data rather than operator measurements. Multi-cell context does not yet imply coordinated multi-cell RL, and the PPO environment remains single-step. The energy-efficiency index is a normalised simulator quantity, not a bits-per-joule measurement, and an energy reduction under one operating point could in principle trade against coverage, capacity, accessibility, retainability, or quality at another; the reward mitigates this by retaining throughput, fairness, and cell-edge terms, but it does not remove the risk. The retained KPI percentages are deterministic model reference outputs and require a fresh v17.8 neural-PPO validation campaign before they can be presented as new implementation results. Recommended validation includes paired AI-OFF and PPO cases, multiple independent seeds with confidence intervals, held-out load, mobility, and CA configurations, carrier-trace train and test separation, checkpoint compatibility checks, and action-sensitivity tests.

VIII. CONCLUSION

RANOpt AI Beta v17.8 changes the learning mechanism materially. The two 64-node hidden layers now participate directly in training: the policy head produces action probabilities, the value head supplies a learned baseline, the one-step reward forms the advantage, the old and new action probabilities form the PPO ratio, the clipped surrogate objective constrains excessive policy change, and backpropagation carries the gradients through both hidden layers so their weights are actually modified. A dimensionless energy-efficiency index enters the reward with an interpretation that stops short of a physical bits-per-joule claim. The implementation is therefore appropriately described as an explainable neural single-step RL-PPO framework, distinct from a future multi-step MDP, and the retained KPI reference point motivates but does not substitute for the multi-seed validation campaign identified above.

APPENDIX A - STEP-BY-STEP v17.8 PPO TRAINING CYCLE

The training cycle is: (1) build and normalise the 11-feature state vector s ; (2) forward-propagate s through the two 64-node hidden layers; (3) compute the five softmax policy probabilities and $V(s)$; (4) select and evaluate one RRM action and obtain reward r ; (5) compute $A = r - V(s)$; (6) retain the old action probability and form $\rho = \pi_{\text{new}}/\pi_{\text{old}}$; (7) form the clipped surrogate policy objective and the value loss; (8) backpropagate

gradients through the policy and value heads and both hidden layers; (9) apply optimiser updates to weights and biases; and (10) persist the compatible neural checkpoint when enabled.

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